



More complex linear equations

What to do

- Answer the following questions for each of these pairs of points.
 - $(-1, 2)$ and $(1, -2)$
 - $(1, 4)$ and $(-1, 6)$
 - $(2, 3)$ and $(-2, 3)$
 - On a set of Cartesian axes, plot the points and draw a line through them.
 - Find the gradient of the line.
 - Find the y -intercept of the line.
 - Use your answers from **ii** and **iii** to write an equation for the line.
 - Find one more point that lies on the line.
- Find the equation of the line containing the given pairs of points.
 - $(0, -2)$ and $(2, 6)$
 - $(-1, 5)$ and $(-4, 2)$
 - $(-5, -3)$ and $(1, -1)$
 - $(5, -2)$ and $(-2, 1)$
 - $(-5, 5)$ and $(-5, 7)$
 - $(2, -3)$ and $(-4, -3)$
- Graph a line with a gradient of $\frac{1}{4}$ and passing through $(1, 3)$.
 - Write an equation for the line.
 - On the same set of axes, graph another line with a gradient of $\frac{2}{8}$ and passing through $(-2, -1)$.
 - Write an equation for the line.
 - What do you observe about the two equations you have established?
 - What is the name given to these two lines?
- Does the graph of the straight line with a slope of -2 and a y -intercept of 5 pass through the point $(3, -1)$?
- MC** Which one of the following equations contains the two points $(-3, -2)$ and $(3, 4)$?
 - $y = x + 1$
 - $y - 1 = -x$
 - $x + y = -7$
 - $y = -x + 1$
- Find the equation of the line depicted in each of the following graphs.
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- If points lie on the same straight line they are called collinear.
 - Show why the points $(-2, 5)$, $(-1, 3)$ and $(-3, 7)$ are collinear.
 - Find the equation of the line.
 - What is the x -intercept?